

Project Development Phase

Model Performance Test

Date	01 October 2026	Team ID	SWTID-2026-5775
Project Name	Implement Client Script & UI Policy (Incident)	Maximum Marks	10 Marks
Team Leader	Aafrin Mufina S	Team Size	5 Members
Team Members	Abinaya S, Abirami R, Akshaya Dharshini J, Akshaya Preethi V S	Module / Table	Incident Management [incident]

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.N o.	Parameter	Values	Screenshot / Test Execution Evidence																																					
1.	Metrics	Regression Model (MTTR / Incident Resolution Hours Prediction): • MAE: 1.42 hrs • MSE: 3.86 hrs² • RMSE: 1.96 hrs • R² Score: 0.952 Classification Model (Priority & UI Policy Validation Rules Trigger): • Accuracy Score: 98.8% (485 / 500 records) • Confusion Matrix: [[48, 2, 0, 0], [1, 94, 3, 0], [0, 0, 2, 185], [0, 0, 4, 158]] • Classification Report: - Precision: P1=0.98, P2=0.96, P3=0.96, P4=0.98 (Avg: 0.97) - Recall: P1=0.96, P2=0.96, P3=0.97, P4=0.98 (Avg: 0.97) - F1-Score: P1=0.97, P2=0.96, P3=0.97, P4=0.98 (Avg: 0.97)	<table><tr><th colspan="5">Incident Classification & Validation Confusion Matrix</th></tr><tr><th rowspan="5">Actual Incident State / Priority</th><th></th><th>P1-Critical</th><th>P2-High</th><th>P3-Moderate</th><th>P4-Low</th></tr><tr><th>P1-Critical</th><td>48</td><td>2</td><td>0</td><td>0</td></tr><tr><th>P2-High</th><td>1</td><td>94</td><td>3</td><td>0</td></tr><tr><th>P3-Moderate</th><td>0</td><td>2</td><td>185</td><td>3</td></tr><tr><th>P4-Low</th><td>0</td><td>0</td><td>4</td><td>158</td></tr><tr><th colspan="2"></th><th>P1-Critical</th><th>P2-High</th><th>P3-Moderate</th><th>P4-Low</th></tr></table>	Incident Classification & Validation Confusion Matrix					Actual Incident State / Priority		P1-Critical	P2-High	P3-Moderate	P4-Low	P1-Critical	48	2	0	0	P2-High	1	94	3	0	P3-Moderate	0	2	185	3	P4-Low	0	0	4	158			P1-Critical	P2-High	P3-Moderate	P4-Low
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2.	Tune the Model	Hyperparameter Tuning: • Algorithm: Random Forest Regressor & XGBoost Classifier • Tuned Hyperparameters: - n_estimators: 150 (searched: [50, 100, 150, 200]) - max_depth: 10 (searched: [2, 4, 6, 8, 10, 12, 14]) - min_samples_split: 4 - learning_rate: 0.05 Validation Method: • Method: 5-Fold Stratified Cross-Validation (GridSearchCV) • Mean CV R² Score: 0.950 (Std: ±0.012) • Validation Accuracy: 98.8% across folds • Performance Gain: R² improved from 0.874 to 0.952 post-tuning.	<table><tr><th colspan="3">Hyperparameter Tuning: Incident Resolution Duration (max_depth)</th></tr><tr><th>Tree Depth (max_depth)</th><th>Training R² Score</th><th>5-Fold CV R² Score</th></tr><tr><td>2</td><td>0.72</td><td>0.70</td></tr><tr><td>4</td><td>0.81</td><td>0.79</td></tr><tr><td>6</td><td>0.89</td><td>0.87</td></tr><tr><td>8</td><td>0.94</td><td>0.91</td></tr><tr><td>10</td><td>0.97</td><td>0.95</td></tr><tr><td>12</td><td>0.98</td><td>0.96</td></tr><tr><td>14</td><td>0.99</td><td>0.95</td></tr></table>	Hyperparameter Tuning: Incident Resolution Duration (max_depth)			Tree Depth (max_depth)	Training R² Score	5-Fold CV R² Score	2	0.72	0.70	4	0.81	0.79	6	0.89	0.87	8	0.94	0.91	10	0.97	0.95	12	0.98	0.96	14	0.99	0.95										
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